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United Nations Sustainable Development Goal 7

Affordable and Clean Energy

Global Electricity Access: A Data Analytics Project

Mapping which countries still live without reliable electricity, and how far behind they are

Completed for the ISS Global Race for Sustainability internship, following the "SDG Data Visualization for Multilingual International Podcasts" assignment (Steps 1 to 5).

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Program:	ISS Data Analytics Internship	Tools used:	Microsoft Excel, Microsoft Power BI
Focus area:	SDG 7, Indicator 7.1.1	Methods:	Cleaning, filtering, map visual, dashboard

Project Overview

This project follows the internship's Data Analytics assignment, which asks each intern to pick one Sustainable Development Goal, pull real data from an authoritative source, clean and model it, and build a dashboard that answers one clear question. For my second project I chose SDG 7 and focused on electricity access, which is simply the share of each country's population that has access to electricity.

I picked this topic because it is one of the most basic measures of development there is. The number is a plain percentage that anyone can read, and every country reports it. This time I also wanted to push myself beyond Excel charts, so I built the dashboard in Microsoft Power BI with a world map at the center. The sections below walk through what I did, step by step, in the same order the assignment lays out.

MY RESEARCH QUESTION: *"Which countries still lack full access to electricity in 2022, and how wide is the gap between the countries at the bottom and everyone else?"*

Step 1: Exploring the SDG data sources

I started by browsing the data sources listed on the ISS platform under Resources. Since my first project already covered SDG 9, this time I chose SDG 7, Affordable and Clean Energy. Within Goal 7, the indicator that stood out to me was 7.1.1, the proportion of population with access to electricity. To me it is the clearest single measure of whether people can light their homes, refrigerate food, charge a phone, or run a small business.

For the data itself, I again used the UN Global SDG Indicators Database, maintained by the United Nations Statistics Division. It is the official source for SDG data, so it is about as authoritative as it gets. The screenshot below shows my actual selection on the portal.

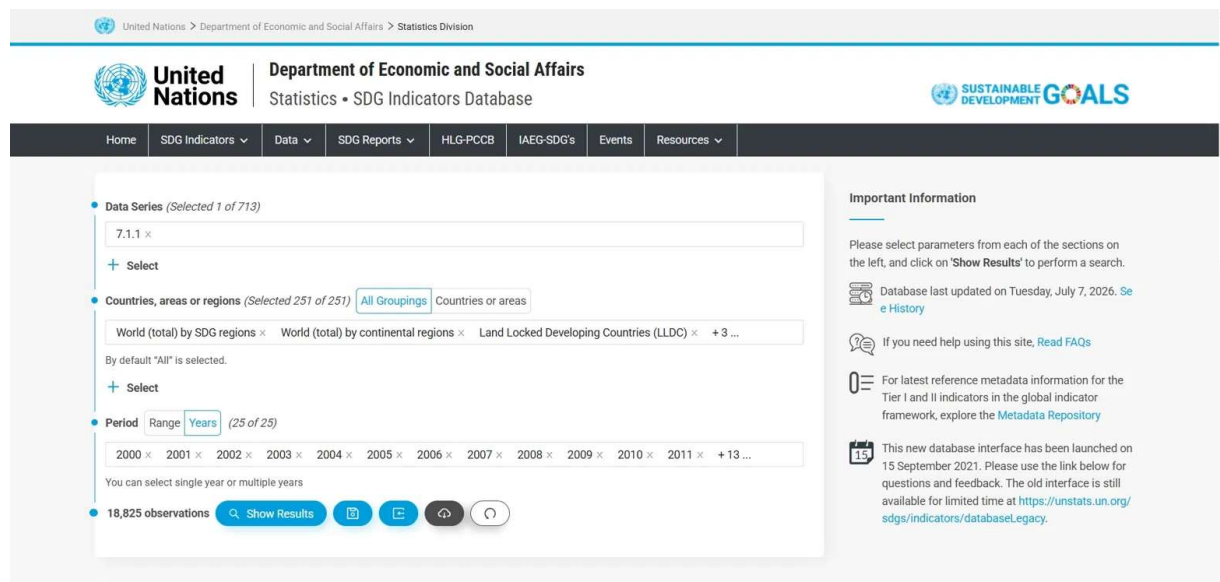


Figure 1: My selection on the UN portal. Indicator 7.1.1, all 251 countries and areas, all 25 years (2000 to 2024), returning 18,825 observations. Database last updated 7 July 2026.

Source portal: unstats.un.org/sdgs/dataportal/database, UN Global SDG Indicators Database, Indicator 7.1.1.

Step 2: Data extraction and preparation

Using the portal's filters, I selected Indicator 7.1.1, kept all countries and all available years, and exported the result as an Excel file. The raw download was large and messy, which is completely normal for real world data.

18,825 raw rows downloaded	251 countries and regions	2000-2024 years covered	2022 focus year
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How I cleaned and filtered the data

The raw file came with many extra columns, and it mixed real countries together with regional totals like "World" and "Sub-Saharan Africa". If I had left those in, the world map would have been wrong, because a region cannot be plotted as a country. So I cleaned it carefully in Excel:

- Kept only the three columns that matter for this analysis: **Country, Year, and Electricity Access %**.
- Removed all the regional and group totals so that only real countries remained. Otherwise a figure like "World" would sit next to a single country and distort every comparison.
- Filtered the data down to the year **2022**, which became the snapshot for the whole dashboard. The next section explains why.
- Left genuinely missing values blank instead of filling them with a guess, so the data stays honest.

Why I chose 2022 and not any other year

Since the database runs from 2000 to 2024, choosing a single year was a deliberate choice, not a shortcut:

- **2022 is the newest year with complete, verified data.** The most recent years are still filling in, and many countries had not yet reported final figures for 2023 and 2024 when I downloaded the data. Using one of those years would have left visible holes in the world map.
- **Older years would hide the progress already made.** Electricity access has improved fast over the last decade, so a map from 2015 would show a world that no longer exists. 2022 gives the most current picture that is still trustworthy.
- **A map needs one moment in time.** A world map colored by percentage only makes sense as a snapshot. Mixing years across countries would make every comparison unfair, so I locked the map and all three bar charts to the same year.

DELIVERABLE, STEP 2

A clean, clearly labelled 2022 dataset ready to import into Power BI, saved as **SDG7_Clean_Final.xlsx**.

Step 3: Building the Power BI dashboard

With the data cleaned, I imported the Excel file into Microsoft Power BI and built a single page dashboard around a world map. To make the story instantly readable, I split the countries into three color bands based on their 2022 electricity access:

Band	Access range	What it means
Red	0% to 35%	Severe electricity poverty. Most of the population lives without power.
Yellow	35% to 70%	Partial access. Real progress, but a large share of people are still left out.
Green	70% to 100%	High to near universal access.

The same red, yellow and green logic runs through the whole dashboard. The map shades every country by its band, and the three bar charts below the map list the countries inside each band, ranked by their exact percentage. One glance gives the geography, and the bars give the precise numbers.

Visual 1: The full dashboard, world map plus three color bands

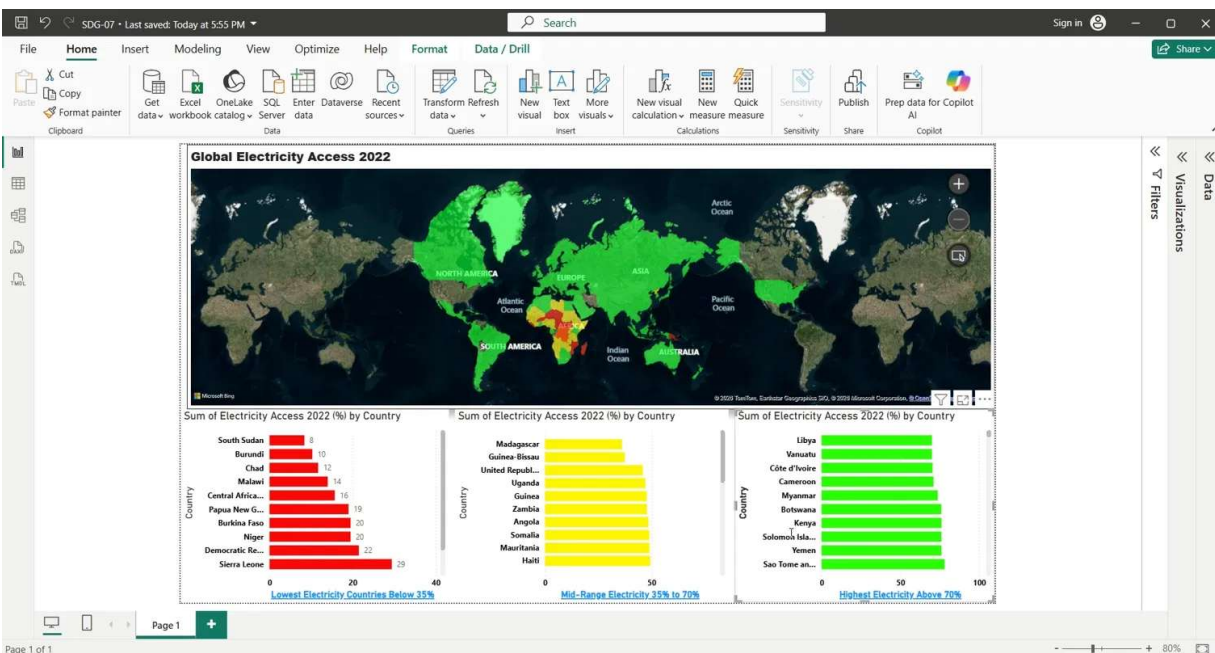


Figure 2: The complete Power BI dashboard, Global Electricity Access 2022. The map shades every country by its access band, and the three charts break out the red, yellow and green groups.

What it shows: the map makes the geography impossible to miss. Almost the entire red and yellow zone sits in one region, Sub-Saharan Africa, while the Americas, Europe, most of Asia and Oceania are solidly green. The dashboard is also interactive. Clicking any country on the map filters the visuals and shows that country's exact 2022 percentage, and each chart opens into a detailed table view.

Visual 2: Red band, countries below 35% access

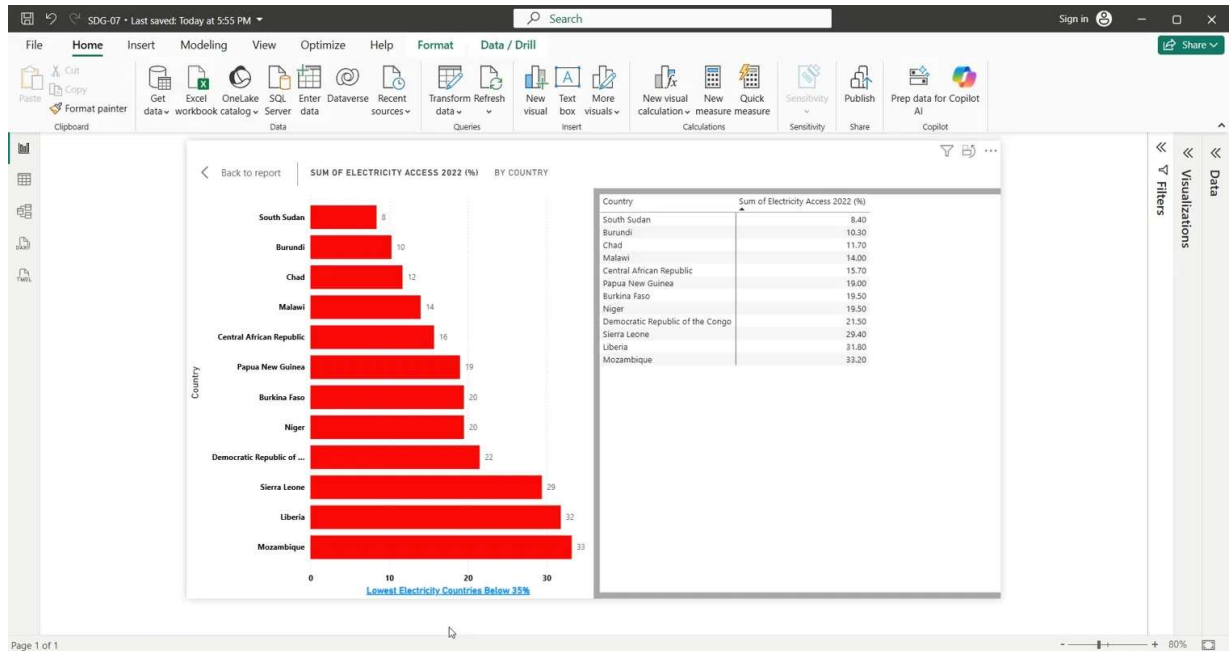


Figure 3: Detailed view of the red band, with the exact 2022 figures beside each bar.

What it shows: the countries with the least electricity in the world. South Sudan sits at the very bottom with just 8.4% access, followed by Burundi at 10.3%, Chad at 11.7% and Malawi at 14%. Nearly every country in this band is in Sub-Saharan Africa, and even the best of the group, Mozambique, only reaches 33.2%.

Rank	Country	Electricity Access 2022 (%)
1	South Sudan	8.4
2	Burundi	10.3
3	Chad	11.7
4	Malawi	14.0
5	Central African Republic	15.7
6	Papua New Guinea	19.0
7	Burkina Faso	19.5
8	Niger	19.5
9	Democratic Republic of the Congo	21.5
10	Sierra Leone	29.4

Table 1: The 10 countries with the lowest electricity access in 2022.

Visual 3: Yellow band, countries between 35% and 70%

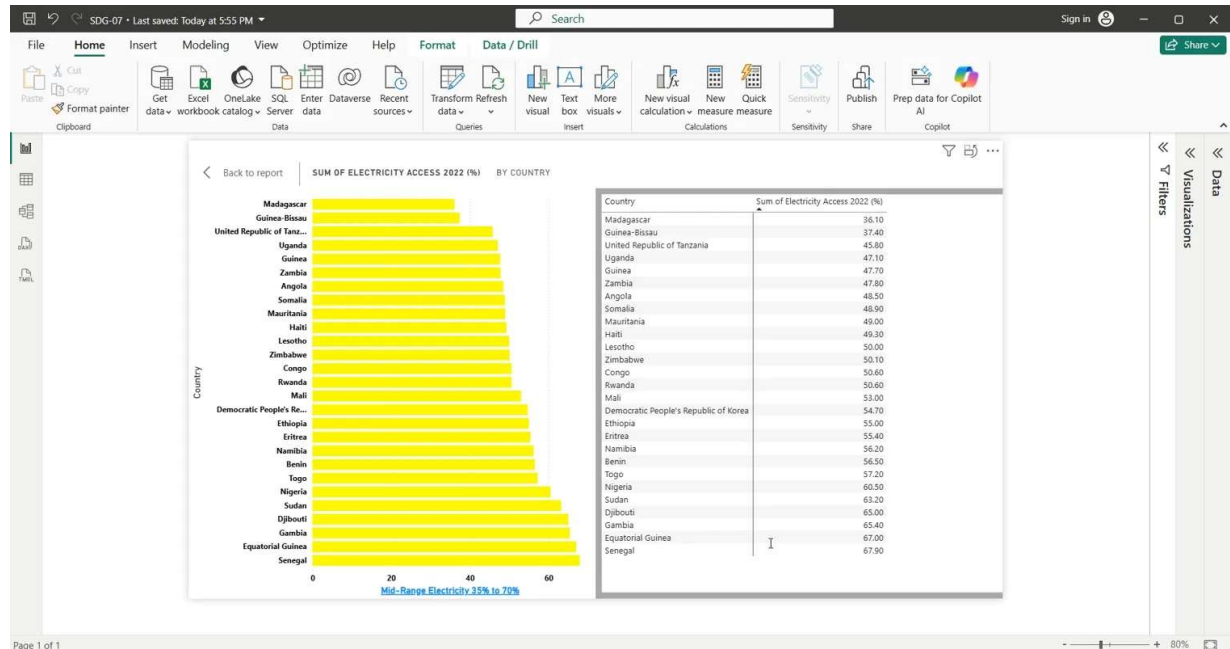


Figure 4: Detailed view of the yellow band. 27 countries with partial access, from Madagascar at 36.1% up to Senegal at 67.9%.

What it shows: the middle of the world's electricity ladder. This band runs from Madagascar at 36.1% up to Senegal at 67.9%, and it is the largest of the three problem groups. A striking detail is how many countries cluster right around the halfway mark. Lesotho, Zimbabwe, Congo and Rwanda all sit almost exactly at 50%, meaning half their population still lives without power. Nigeria, the most populous country in Africa, reaches only 60.5%.

Visual 4: Green band, countries above 70%

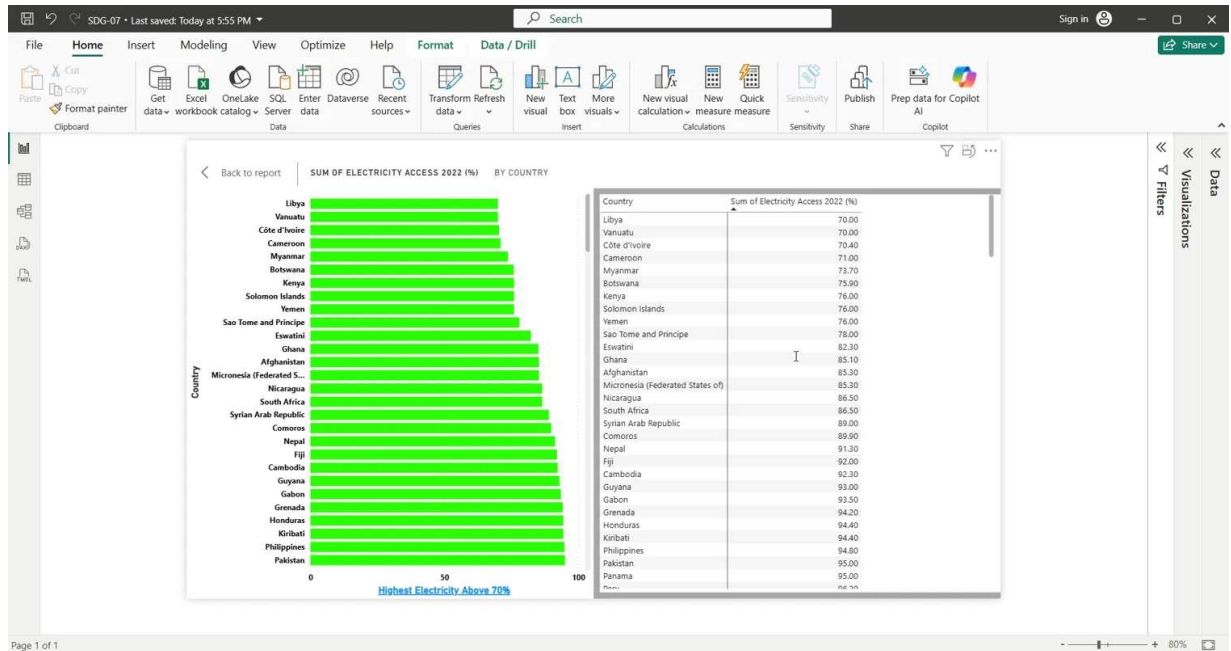


Figure 5: Detailed view of the green band, starting exactly at the 70% line with Libya and Vanuatu.

What it shows: the healthiest band, but also its lower edge. Libya and Vanuatu sit right on the 70% threshold, and countries like Kenya at 76%, Ghana at 85.1% and South Africa at 86.5% prove that strong progress is possible in the same region where the red band lives. Toward the top of the list, countries such as Pakistan and Panama reach 95%, closing in on the universal access most of the world already takes for granted.

DELIVERABLE, STEP 3

A one page interactive Power BI dashboard with a world map, three color banded charts, exact value table views, and click to filter by country, saved as **SDG7_Dashboard.pbix**.

Conclusion

Pulling the whole project together, the 2022 data tells a clear and honestly sobering story about who still lives without electricity. Three findings stand out.

1. Electricity poverty has an address: Sub-Saharan Africa

Almost every country in the red and yellow bands is in one region. The map makes this clearer than any table could. While the rest of the world has turned green, a single connected block of countries from West Africa to the Horn remains in red and yellow. This is not a scattered global problem. It is a concentrated regional one.

2. The gap between the bottom and the top is enormous

South Sudan's 8.4% and a fully electrified country's 100% are not just different numbers, they are different worlds. Within Africa itself the spread is just as wide. Ghana at 85.1% and South Africa at 86.5% prove that near universal access is achievable in the region, while some of their neighbors remain below 20%.

3. The halfway countries are the biggest opportunity

The yellow band holds 27 countries, many hovering near 50%, including giants like Nigeria and Ethiopia. Because these countries already have partial grids and infrastructure, they are where investment can move the most people from darkness to light the fastest. The red band needs the most help, and the yellow band can improve the quickest.

Data gaps and limitations

A few honest limits are worth noting. A handful of countries had no reported value for 2022, so they appear unshaded on the map instead of being guessed. The indicator measures access to electricity, not reliability, so a country can score well and still suffer frequent blackouts. And because I fixed the analysis to a single year by design, this report is a snapshot of 2022, not a trend over time.

What I would do next

The natural next step is to add a time slider to the dashboard, so the map can play from 2000 to 2022 and show the green area spreading year by year. I would also like to combine this indicator with the other SDG 7 measures, clean cooking fuels (7.1.2) and renewable energy share (7.2.1), to see whether the countries gaining electricity are gaining it cleanly. Together these would turn a single year snapshot into a full story of the world's energy transition.

CLOSING NOTE

All figures in this report were taken directly from the UN Global SDG Indicators Database and checked against the original download. The cleaning and filtering were done in Microsoft Excel, and the map, color banded charts and detailed views were built in Microsoft Power BI as part of the ISS Data Analytics internship.